

Analysis First-Year Exam, May 2020, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $(a_n)_{n=0}^{\infty}$ be a bounded real sequence. Write L for the set of all real numbers ℓ such that $(\exists N)(\forall n > N) \ell < a_n$ and write U for the set of all real numbers u such that $(\exists N)(\forall n > N) a_n < u$.
 - (i) Prove that $\sup L \leq \inf U$.
 - (ii) Give an example of a convergent sequence for which the inequality in (i) is strict, or prove that no such sequence exists.
2. Let U be a subset of the metric space X . Does either of the following conditions imply the other? Prove or disprove, as appropriate.
 - (i) U is open.
 - (ii) For any $A \subseteq X$, if U meets the closure of A then U meets A itself.
3. Let X and Y be metric spaces. Decide whether the following statements are true or false.
 - (i) If $(x_n)_{n=0}^{\infty}$ converges to x in X then $\{x\} \cup \{x_n : n \geq 0\}$ is compact.
 - (ii) The function $f : X \rightarrow Y$ is continuous if its restriction $f|_K$ to each compact subset $K \subseteq X$ is continuous.
4. Let $f : X \rightarrow Y$ be a continuous bijection with inverse $g : Y \rightarrow X$ and assume that X is complete.
 - (i) Show by example that if g is continuous then Y need not be complete.
 - (ii) Prove that if g is uniformly continuous then Y must be complete.
5. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be differentiable and assume $\lim_{x \rightarrow \infty} f'(x) = L \in \mathbb{R}$.
 - (i) When a is a positive real number, calculate
$$\lim_{x \rightarrow \infty} \frac{f(x+a) - f(x)}{a}.$$
 - (ii) Assume further that $f(x) \rightarrow A \in \mathbb{R}$ as $x \rightarrow \infty$ and calculate L .